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ACM/IDS 104 - Problem Set 5 - MATLAB Problems

Before writing your MATLAB code, it is always good practice to get rid of any leftover variables and figures from previous scripts.

```
clc; clear; close all;
```

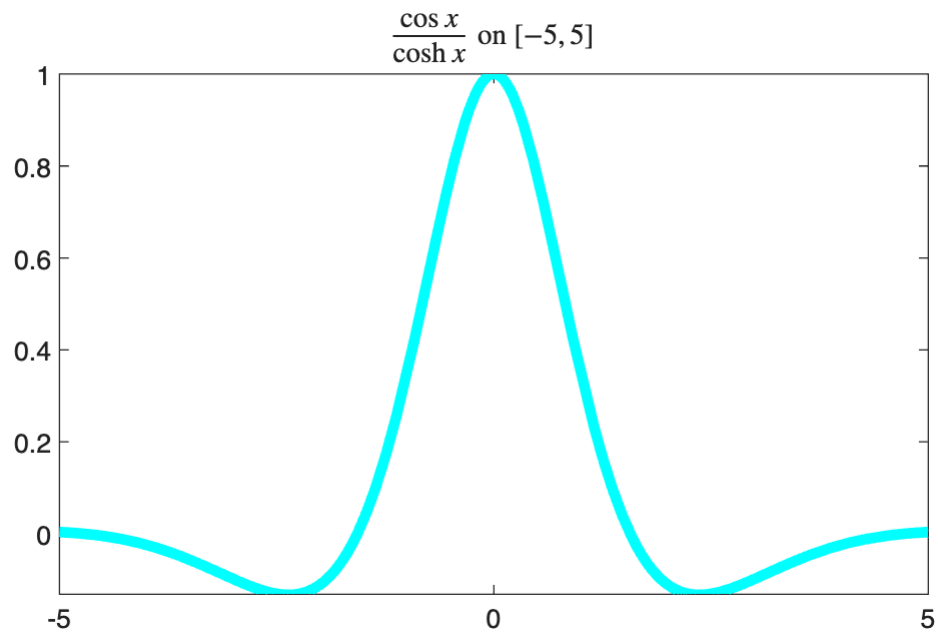
Problem 1 (10 points) Application of Projections to Approximation

In Problem 4 of PS4, we saw that even higher degree interpolating polynomials may not be accurate approximations to complex functions. We have the function:

$$f(x) = \frac{\cos x}{\cosh x}, \quad \text{on } [-a, a], \quad a = 5$$

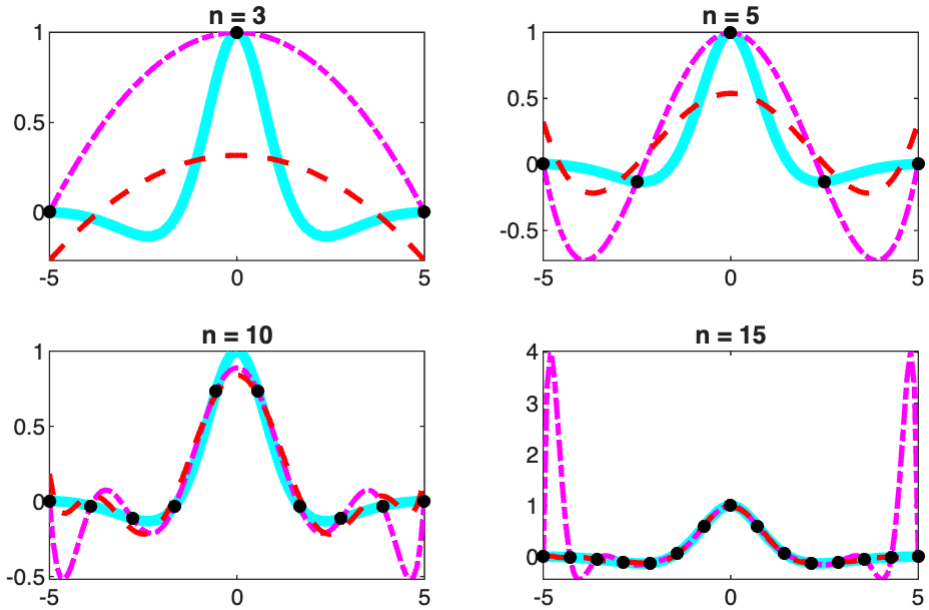
Let us recall how this function looks like and how its interpolating polynomials of degree $(n-1)$ for $n = 3, 5, 10, 15$ behave:

```
%{
Setup
%}
f = @(x) cos(x)./cosh(x); % our function
a = 5; % setting the value of a
n = [3 5 10 15]; % setting the number of points
sub = 1; % subplot index

%{
How f(x) looks like on [-5, 5]
%}
figure;
fplot(f, [-a, a], "-c", "lineWidth", 4);
title("$\frac{\cos{x}}{\cosh{x}}$ on $[-5, 5]$","Interpreter","latex");
```




```
%{
Read the discussion below and complete the code
%}
figure;
for ival = a
    for degree = n-1
        %{
        INTERPOLATING POLYNOMIALS -- no changes needed
        -> Select degree+1 points in the interval
        -> Evaluate f(x) on these points
        -> Find the polynomial coefficients
        %}
        pts = ones(degree+1, 2); % initializing the points
        pts(:, 1) = linspace(-ival, ival, degree+1); % setting the x-values
        for i = 1 : degree+1
            pts(i, 2) = f(pts(i, 1)); % evaluating cos(x) /
cosh(x)
        end
        coeffs = polyfit(pts(:, 1), pts(:, 2), degree); % coefficients
        %{
        ORTHOGONAL PROJECTIONS -- TODO
        -> Get transformed Legendre polynomials
        -> Find alpha_k using L^2 inner product
        -> Evaluate alpha_k*Q_k
        %}
        alpha = zeros(degree+1, 1);
        for k = 0:degree
            % legendre
            Qk = @(x) legendreP(k, x/ival);
```

```
            % <f, Q_k>
            num = integral(@(x) f(x).*Qk(x), -ival, ival);

            % <Q_k, Q_k>
            den = integral(@(x) Qk(x).^2, -ival, ival);

            alpha(k+1) = num / den;
        end

        %{
        PLOTTING
        Plot f(x), the sampled points, interpolating and approximating
        polynomials
        Please use different colors and linestyles
        %}
        subplot(2, 2, sub);
        fplot(f, [-ival, ival], "-c", "lineWidth", 4);
        hold on
        interpoints = linspace(-ival, ival);
        p = polyval(coeffs, interpoints); % evaluating coeffs in interval
        plot(interpoints, p, "-.m", "lineWidth", 2);

        p_approx = zeros(size(interpoints));
        for k = 0:degree
            Qk_vals = legendreP(k, interpoints/ival);
            p_approx = p_approx + alpha(k+1)*Qk_vals;
        end
        plot(interpoints, p_approx, "--r", "lineWidth", 2);

        plot(pts(:, 1), pts(:, 2), "ok", "MarkerSize", 2, "lineWidth", 3);
        title(strcat("n = ", int2str(degree+1)));
        sub = sub + 1; % increase subplot index
    end
end
```



Now, instead of interpolating polynomials, let us approximate $f(x)$ by its orthogonal projection onto the inner space $\mathscr{P}^{(n-1)}_{[-a,a]}$ of polynomials on $[-a,a]$, equipped with the L^2 inner product:

$$f(x) \approx p(x) = \mathrm{pr}_{\mathscr{P}^{(n-1)}_{[-a,a]}} f(x)$$

Recall (Lecture 10) that $p(x)$ is the closest (in the L^2 sense) polynomial to $f(x)$ in $\mathscr{P}^{(n-1)}_{[-a,a]}$, i.e.

$$p(x) = \arg \min_{q \in \mathscr{P}^{(n-1)}_{[-a,a]}} \|f(x) - q(x)\|$$

We know that the transformed Legendre polynomials $\widetilde{Q}_0(x), \cdots, \widetilde{Q}_{n-1}(x)$ form an orthogonal basis of $\mathscr{P}^{(n-1)}_{[-a,a]}$, and, therefore:

$$p(x) = \sum_{k=0}^{n-1} \alpha_k \widetilde{Q}_k(x)$$

where α_k are the coordinates of $p(x)$ in that basis.

Modify the above code to find the approximating polynomials as well. Plot each approximating polynomial on its corresponding subplot. Useful functions for this problem:

`legendreP(), integral()`

ACM 104 HW #5

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2. Hermite Polynomials

a) $\{v_0, v_1, v_2\} = \{1, x, x^2\}$

$h_0(x) = v_0 = 1$

$h_1(x) = v_1 - \dfrac{\langle v_1, h_0\rangle}{\langle h_0, h_0\rangle} h_0 = x$

$\langle h_0, h_0\rangle = \int_{-\infty}^{\infty} e^{-x^2/2}\,dx = \sqrt{2\pi}$

$\langle v_1, h_0\rangle = \int_{-\infty}^{\infty} x e^{-x^2/2}\,dx = 0$ (odd)

$h_2(x) = v_2 - \dfrac{\langle v_2, h_0\rangle}{\langle h_0, h_0\rangle} h_0 - \dfrac{\langle v_2, h_1\rangle}{\langle h_1, h_1\rangle}$

$\langle v_2, h_0\rangle = \langle x^2, 1\rangle = \int_{-\infty}^{\infty} x^2 e^{-x^2/2}\,dx = \sqrt{2\pi}.$

$\langle h_1, h_1\rangle = \langle x, x\rangle = \int_{-\infty}^{\infty} x^2 e^{-x^2/2}\,dx = \sqrt{2\pi}$

$\langle v_2, h_1\rangle = \langle x^2, x\rangle = \int_{-\infty}^{\infty} x^3 e^{-x^2/2}\,dx = 0$ (odd)

$$\boxed{h_2(x) = x^2 - 1,\ h_1 = x,\ h_0 = 1}$$

Lemma 1 : We can evaluate $\langle x^{\frac{n}{2}}, x^{\frac{n}{2}}\rangle = \int_{-\infty}^{\infty} x^n e^{-x^2/2}\,dx = I_n$ using recurrence relation and integration by parts.

$u = x^{n-1}$ $\quad du = (n-1)x^{n-2}$

$dv = xe^{-x^2/2}\,dx$ $\quad v = -e^{-x^2/2}$

$\longrightarrow I_n = \Big[\underbrace{-x^{n-1}e^{-x^2/2}}_{0}\Big]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \left(-e^{-x^2/2}\right)(n-1)x^{n-2}\,dx$

$I_n = (n-1)\int_{-\infty}^{\infty} x^{n-2} e^{-x^2/2}\,dx \rightarrow \underline{I_n = (n-1)\,I_{n-2}}$

So, $I_2 = (2-1)I_0 = 1 \cdot \int_{-\infty}^{\infty} e^{-x^2/2}\,dx = \sqrt{2\pi}$

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b) $h_0(x) = 1$: $\|h_0\|^2 = \sqrt{2\pi} \rightarrow \boxed{H_0(x) = \dfrac{1}{\sqrt[4]{2\pi}}}$

$h_1(x) = x$: $\|h_1\|^2 = \sqrt{2\pi} \rightarrow \boxed{H_1(x) = \dfrac{x}{\sqrt[4]{2\pi}}}$

$h_2(x) = x^2$: $\|h_2\|^2 = \langle x^2 - 1, x^2 - 1\rangle = \langle x^2, x^2 - 1\rangle - \langle 1, x^2 - 1\rangle = \langle x^2, x^2 - 1\rangle$

$= \langle x^2, x^2\rangle - \langle x^2, 1\rangle = (4-1)I_2 - I_2 = 2\sqrt{2\pi} \rightarrow \boxed{H_2(x) = \dfrac{x^2 - 1}{\sqrt{2}\,\sqrt[4]{2\pi}}}$

I_4

c) $p(x) = \frac{\langle f, h_0 \rangle}{\|h_0\|^2} h_0(x) + \frac{\langle f, h_1 \rangle}{\|h_1\|^2} h_1(x) + \frac{\langle f, h_2 \rangle}{\|h_2\|^2} h_2(x)$

Compute: $\langle e^x, g(x) \rangle = \int_{-\infty}^{\infty} g(x) e^x e^{-x^2/2} dx$

$e^x e^{-x^2/2} = e^{-1/2(x^2 - 2x)} = \sqrt{e}\, e^{-(x-1)^2/2}$; $u = x-1$, $x = u+1$, $dx = du$

$\langle e^x, g(x) \rangle = \sqrt{e} \int_{-\infty}^{\infty} g(u+1) e^{-u^2/2} du$

Let $I_n = \int_{-\infty}^{\infty} u^n e^{-u^2/2} du$; $I_n = (n-1) I_{n-2}$

$I_0 = \sqrt{2\pi}$, $I_1 = 0$, $I_2 = \sqrt{2\pi}$

$h_0(x) = 1$: $\langle e^x, 1 \rangle = \sqrt{e}\, I_0 = \sqrt{2\pi e} \longrightarrow c_0 = \frac{\sqrt{2\pi e}}{\sqrt{2\pi}} = \sqrt{e}$

$h_1(x) = x \to x = u+1 \to \langle e^x, x \rangle = \sqrt{e} \int_{-\infty}^{\infty} (u+1) e^{-u^2/2} du = \sqrt{e}(I_1 + I_0) = \sqrt{2\pi e}$

$c_1 = \frac{\sqrt{2\pi e}}{\sqrt{2\pi}} = \sqrt{e}$

$h_2(x) = x^2 - 1 \to x = u+1 \to h_2(u+1) = u^2 + 2u$

$\langle e^x, x^2 - 1 \rangle = \sqrt{e} \int_{-\infty}^{\infty} (u^2 + 2u) e^{-u^2/2} du = \sqrt{e}(I_2 + 2I_1) = \sqrt{2\pi e}$

$c_2 = \frac{\sqrt{2\pi e}}{2\sqrt{2\pi}} = \frac{\sqrt{e}}{2}$

$p(x) = \sqrt{e} + x\sqrt{e} + \frac{(x^2-1)\sqrt{e}}{2}$

$$\boxed{p(x) = \frac{\sqrt{e}}{2}(x+1)^2}$$

3. Orthogonal Components

a1) $x = v_1 \times v_2 = \begin{bmatrix} 2 \\ 5 \\ -4 \end{bmatrix}$

$$\boxed{W_1^{\perp} = \text{span}\left\{ \begin{bmatrix} 2 \\ 5 \\ -4 \end{bmatrix} \right\}}$$

a2) $\langle x, v_1 \rangle_2 = 0 \rightarrow x_1(1) + 2x_2(2) + 3x_3(3) = 0 \rightarrow x_1 + 4x_2 + 9x_3 = 0$

$\langle x, v_2 \rangle_2 = 0 \rightarrow x_1(2) + 2x_2(1) + 3x_3(1) = 0 \rightarrow 2x_1 + 3x_3 = 0 \rightarrow x_1 = -\frac{3}{2}x_3$

Substituting, $-\frac{3}{2}x_3 + 4x_2 + 9x_3 = 0 \rightarrow x_2 = -\frac{15}{8}x_3$

Let $x_3 = 8 \rightarrow x_2 = -15 \rightarrow x_1 = -12$

$$\boxed{W_2^{\perp} = \text{span}\left\{ \begin{bmatrix} -12 \\ -15 \\ 8 \end{bmatrix} \right\}}$$

b) $\langle A, B \rangle = tr(A^T B)$

$\langle A, I_n \rangle = tr(A^T I_n) = tr(A^T) = tr(A)$

The condition for orthogonality is $tr(A) = 0$

U^T is the subspace of all matrices w/ $tr(A) = 0$. Formally:

$$\boxed{U^{\perp} = \{ A \in M_{m \times n} \mid tr(A) = 0 \}}$$

c) Standard basis: $\{1, x, x^2, x^3\}$

Find $p(x) \in P^{(3)} \ni \langle p, x \rangle = 0 \wedge \langle p, x^3 \rangle = 0$

Basis vectors x and x^2 are odd.

For any even $p(x)$, the inner products will be 0 since $\int_{-n}^{n} \text{odd} \cdot \text{even} = \int_{-n}^{n} \text{odd} = 0$

Example: $h(x) = 1 \rightarrow \langle 1, x \rangle = \int_{-1}^{1} x\,dx = 0$, $\langle 1, x^3 \rangle = \int_{-1}^{1} x^3\,dx = 0$

$\langle x^2, x \rangle = \int_{-1}^{1} x^3\,dx = 0$, $\langle x^2, x^3 \rangle = \int_{-1}^{1} x^5\,dx = 0$ $\rightarrow$ So $x^2 \in V^{\perp}$

But could $p(x)$ have any odd terms? Let $p(x) = ax + bx^3$

$\langle p, x \rangle = \int_{-1}^{1} (ax^2 + bx^4)\,dx = 2\left(\frac{a}{3} + \frac{b}{5}\right)$

$\langle p, x^2 \rangle = \int_{-1}^{1} (ax^4 + bx^6)\,dx = 2\left(\frac{a}{5} + \frac{b}{7}\right)$

For $p \in V^{\perp}$, both $= 0$. Det system $= \frac{1}{21} - \frac{1}{25} \neq 0$ so only trivial solution exists.

Therefore, $V^{\perp}$ has no odd components.

4. Real and Complex Eigenbases

a) $A - \lambda I = \begin{bmatrix} -\lambda & 0 & -1 \\ 0 & 1-\lambda & 0 \\ 1 & 0 & -\lambda \end{bmatrix} \longrightarrow \det(A-\lambda I) = (1-\lambda)(\lambda^2+1)$

$\lambda_1 = 1,\ \lambda_2 = i,\ \lambda_3 = -i$

A is $\boxed{\text{complete}}$ as its a 3x3 matrix w/ 3 distinct, linearly dependent eigenvalues.

b) No, because it has complex eigenvalues with only one real one.

c) Yes, since it has 3 distinct, linearly dependent eigenvalues.

$\lambda_1 = 1$: $(A-I)v = 0 \rightarrow \begin{bmatrix} -1 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & -1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ $\begin{matrix} x - z = 0 \rightarrow x = z \\ -x - z = 0 \\ x = 0,\ z = 0 \\ \text{Let } y = 1 \end{matrix}$, $v_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

$\lambda_2 = i$: $(A - iI)v = 0 \rightarrow \begin{bmatrix} -i & 0 & -1 \\ 0 & 1-i & 0 \\ 1 & 0 & -i \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ $\begin{matrix} (1-i)y = 0 \rightarrow y = 0 \\ -ix - z = 0 \rightarrow z = -ix \\ x - iz = x - i(-ix) \rightarrow x - x = 0 \checkmark \\ \text{Let } x = 1 \rightarrow z = -i \end{matrix}$, $v_2 = \begin{bmatrix} 1 \\ 0 \\ -i \end{bmatrix}$

$\lambda_3 = -i$: $(A + iI)v = 0 \rightarrow$ Since A is real, $\overline{v_2} = v_3 = \begin{bmatrix} 1 \\ 0 \\ i \end{bmatrix}$ $\boxed{\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -i \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ i \end{bmatrix} \right\}}$

d.

$u = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix},\ v = \begin{bmatrix} 1 \\ 0 \\ -i \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + i\begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} \longrightarrow B = \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} \right\}$

clearly a basis for $\mathbb{R}^3$ (same as original w/ sign flip/reordering) ↗

$Ax = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \doteq \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

Since $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ is not a scalar multiple of $x = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$,

x is not an eigenvector and

B is not an eigenbasis.

5. Gershgorin Theorem

a) $r1$: center $c_1 = b_{11} = 0$

radius $R_1 = |b_{12}| + |b_{13}| = 1 + 0 = 1$

Disk $D_1 = \{z \in \mathbb{C} : |z| \leq 1\}$

$r2$: center $c_2 = b_{22} = 1$

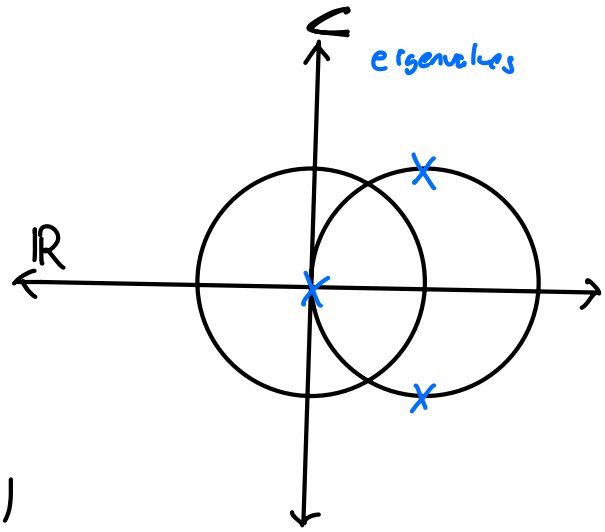
radius $R_2 = |b_{21}| + |b_{23}| = 0 + 1 = 1$

Disk $D_2 = \{z \in \mathbb{C} : |z-1| \leq 1\}$

$r3$: center $c_3 = b_{33} = 1$

radius $R_3 = |b_{31}| + |b_{32}| = 0 + 1 = 1$

Disk $D_3 = \{z \in \mathbb{C} : |z-1| \leq 1\} = D_2$



$$\boxed{D_B : D_1 \cup D_2 \cup D_3 = \{z : |z| \leq 1\} \cup \{z : |z-1| \leq 1\}}$$

b) By Gershgorin Theorem, some eigenvalue λ must lie in the union of the two disks of $A \rightarrow \lambda \in D_A$.

Because $\det(A - \lambda I) = \det((A - \lambda I)^T) = \det(A^T - \lambda I)$, we can apply Gershgorin to $A^T \rightarrow \lambda$ must also lie in the union of those disks $\longrightarrow \lambda \in D_{A^T}$

Therefore: $\boxed{\lambda \in D_A \cap D_{A^T} = D_A^*}$

c) For B^T

r_1: $c_1' = 0$

$R_1' = 0 + 0 = 0$

$D_1' = \{0\}$

r_2: $c_2' = 1$

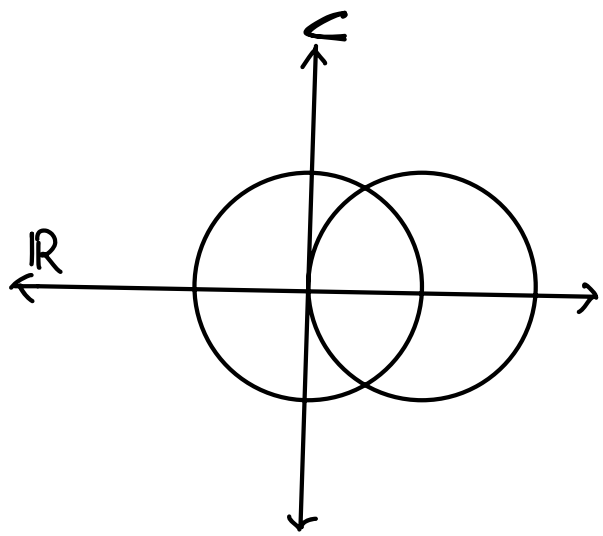
$R_2' = 1 + 1 = 2$

$D_2' = \{z : |z-1| \leq 2\}$

r_3: $c_3' = 1$

$R_3' = 0 + 1 = 1$

$D_3' = \{z : |z-1| \leq 1\}$



$D_{B^T} : D_1' \cup D_2' \cup D_3' = \{z : |z-1| \leq 2\}$

$D_B^* = D_B \cup D_{B^T} = \boxed{D_B = D_B^*}$ since $D_B \subset D_{B^T}$

d) $\det(B - \lambda I) = 0$

$$\begin{vmatrix} -\lambda & 1 & 0 \\ 0 & 1-\lambda & 1 \\ 0 & -1 & 1-\lambda \end{vmatrix} = 0 \rightarrow -\lambda \begin{vmatrix} 1-\lambda & 1 \\ -1 & 1-\lambda \end{vmatrix} = 0 \rightarrow -\lambda(\lambda^2 - 2\lambda + 2) = 0$$

$$\lambda_1 = 0, \ \lambda_2 = 1+i, \ \lambda_3 = 1-i$$

λ_1 is at center inside D_B^*

λ_2 and λ_3 are at the marked x's in the diagram at the edge of the intersection of disks inside D_B^*. ✓

e) $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$, $\det(A) = 1 \neq 0$ so A is nonsingular.

Eigenvalues: 1, 1 (neither is 0)

Gershgorin Disks:

r_1: $C_1 = 1$, $R_1 = 2$, $D_1 = \{z : |z-1| \leq 2\}$

r_2: $C_2 = 1$, $R_2 = 0$, $D_2 = \{1\}$

Union domain contains D_1, which contains 0.

$\boxed{\text{Therefore } A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \text{ is valid counterexample.}}$